ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION
CO3 SCENARIO BASED ASSIGNMENT

Design and Performance Evaluation of an Integrated Photodetection, Imaging, Biomedical Sensing and Display System

Item	Details
Course Outcome	CO3 (Bloom's K3 – Apply, K4 – Analyze)
Maximum Marks	100
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Department / Section	ECE
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Contents: Assumptions · Stage 1 Requirements · Stage 2 Detector selection · Stage 3 Noise and SNR · Stage 4 Speed and bandwidth · Stage 5 PV, image sensors, displays · Stage 6 Applications · Stage 7 MATLAB/SCILAB simulation · Final verification table · Final system design · Final engineering decision · References

Key Points

- **Detector choice:** Hybrid design – Si PIN photodiode (R = 0.60 A/W, η = 79 %) for PPG and an InGaAs APD (R = 0.80 A/W, η = 64 %) for LiDAR and low-light detection.
- **Photocurrents:** PPG $\Delta I = 0.60 \times 25\ \mu W = 15\ \mu A$; LiDAR $I = 0.80 \times 3\ \mu W = 2.4\ \mu A$ (M = 1), raised by APD gain.
- **Noise and SNR:** Given SNR = −22 − (−37) = 15 dB (31.6), which is marginal against the 15.6 dB needed for BER 10^{-9}.
- **APD gain:** The optimum gain is $M \approx 6$, giving 36.6 dB against 28.2 dB at M = 1 (+8.4 dB); above this, excess noise reduces SNR.
- **SNR improvement:** Optical filtering, bandwidth matching, mild cooling and pulse averaging bring the expected SNR to 20 dB or more.
- **Speed:** Transit-time limit f = 0.44/20 ps = 22 GHz; the RC limit is 530 MHz (1 kΩ, 0.3 pF), so RC, not transit time, limits the receiver.
- **Photovoltaic:** P_max = FF × Voc × Isc = 0.75 × 0.60 V × 5 A = 2.25 W; flexible thin-film PV suits wearables.
- **Imaging and display:** Back-side-illuminated CMOS sensor for face recognition and automotive vision; MicroLED for AR/VR/MR and flexible OLED for foldable displays.
- **Applications:** PPG uses a pulsed LED with a TIA and filtering for heart rate and SpO_2; NIR spectroscopy uses the Beer–Lambert relation $A = \varepsilon \cdot c \cdot l$.

0. Given Data and Engineering Assumptions

Parameters not stated in the problem are assumed below, using typical datasheet-level values. All calculations in this report use these values, so the results are reproducible.

Parameter	Value	Source / reason
PPG detector responsivity / ΔP	0.60 A/W / 25 μW	Given
LiDAR responsivity / received power	0.80 A/W (M = 1) / 3 μW	Given
Low-light signal / noise	−22 dBm / −37 dBm	Given
Transit time	20 ps (f ≈ 0.44/tr)	Given
PV	Voc = 0.60 V, Isc = 5 A, FF = 0.75	Given
PPG wavelength	940 nm (NIR LED), Si PIN	Assumed; Si has R ≈ 0.6 A/W near 940 nm
LiDAR wavelength / APD material	1550 nm, InGaAs APD	Assumed; eye-safe, η = 64 % at R = 0.8 A/W
APD dark current / k-factor	Id = 2 nA (multiplied), k = 0.2	Assumed typical InGaAs/InAlAs APD
Receiver load / total capacitance	R_L = 1 kΩ (TIA equivalent), C = 0.3 pF	Assumed
LiDAR receiver bandwidth	B = 500 MHz	Assumed (about 0.7 ns rise time)
PPG front end	R_f = 1 MΩ, B = 100 Hz, Id = 1 nA, A = 1 mm²	Assumed
LiDAR APD active area	200 μm diameter (A = 3.14×10^{-4} cm²)	Assumed
Temperature	T = 300 K	Assumed

Constants: q = 1.602×10^{-19} C, k_B = 1.381×10^{-23} J/K, h = 6.626×10^{-34} J·s, c = 3×10^{8} m/s. Quantum efficiency uses η = R·hc/(qλ) = R × 1240 / λ(nm).

STAGE 1 – Interpret the Application Requirements (15 marks)

The requirements are worked backward from what each application must measure to the detector, sensor and display properties needed.

1.1 Photodetection requirements by application

Topic	Requirement and discussion
Wearable PPG (small intensity changes)	The useful signal is a small AC change (a few percent or less) on a large DC level from skin and tissue. The detector must be linear, low-noise and have good responsivity at 530–940 nm. Moderate speed (below 1 kHz) is enough, so a PIN diode with a high-value TIA works. Ambient-light rejection and motion tolerance matter more than bandwidth.

Topic	Requirement and discussion
Fast automotive LiDAR	Needs ns-scale pulse detection (bandwidth in the hundreds of MHz), high gain for μW returns, low capacitance, and a wide dynamic range from near to far targets. An APD with a low-noise TIA is the usual choice.
Weak-signal detection	Needs high responsivity, high quantum efficiency, low dark current and internal gain, so that the signal exceeds amplifier noise. Noise-equivalent power (NEP) sets the limit.
Responsivity, QE and detectivity	R (A/W) converts optical power to current. QE η is the electrons-per-photon ratio, $R = \eta q\lambda/hc$. Detectivity $D^* = \sqrt{(A \cdot B)}/NEP$ normalises sensitivity for area and bandwidth, so different detectors can be compared fairly.
NEP and low-light capability	NEP is the optical power giving SNR = 1 in 1 Hz bandwidth. A lower NEP means weaker detectable light. Minimum detectable power = $NEP \cdot \sqrt{B}$.
Linearity	Photocurrent must be proportional to power over the whole dynamic range. Linearity is needed for accurate pixel intensity (imaging), PPG amplitude and LiDAR intensity calibration. Saturation, space-charge effects and APD gain compression break linearity.
Response time and bandwidth	The detector must follow the signal: bandwidth ≥ about 0.35/t_rise. Transit time and the RC time constant are the two physical limits.
Transit time vs RC	Transit time is how long carriers take to cross the depletion region (thin region = fast). RC is detector capacitance times load resistance (small area = low C = fast). The two compete, because a thin region raises capacitance.
CMOS and CCD characteristics	High QE, low read noise, low dark current, good full-well capacity, global-shutter option, and low fixed-pattern noise. CMOS additionally needs low power and SoC integration; CCD offers uniform, low-noise charge transfer.
Pixel size, scaling, SoC	Smaller pixels give higher resolution and compact optics, but lower full-well capacity and lower photon collection (lower SNR). Back-side illumination, micro-lenses and stacked designs recover sensitivity. On-chip ADC, ISP and AI blocks (SoC) cut size and power.
Flexible PV	Needs light weight, mechanical flexibility, adequate fill factor, tolerance to bending, and stable output under weak/indoor light.
OLED, MicroLED, quantum-dot displays	Need high brightness, contrast, colour gamut, fast response, long lifetime and low power. OLED is self-emissive with deep blacks; MicroLED is self-emissive inorganic with very high brightness; quantum-dot (QD) layers give narrow-band, pure colours.
AR/VR/MR and foldable displays	AR/VR/MR needs very high pixel density (more than 2000 ppi), high brightness (waveguide losses), fast response (below 1 ms) and low power. Foldable displays need thin, flexible, bend-tolerant emissive layers.

Topic	Requirement and discussion
Practical constraints	Power budget (battery wearable), heat (APD gain drifts with temperature), integration (optics, ICs, flex PCBs), reliability (OLED ageing, APD bias safety) and complexity (cost, calibration).

1.2 Application Requirement Statement

The platform shall (i) resolve a 25 µW intensity change as a clean PPG signal using a low-noise, linear Si PIN receiver; (ii) detect 3 µW LiDAR returns with an APD receiver of at least about 500 MHz bandwidth and SNR above the 15.6 dB needed for a 10^{-9} error rate; (iii) hold a low-light SNR of at least 15 dB, with a target of 20 dB or more; (iv) use detectors whose 20 ps transit time does not limit the receiver (22 GHz capability); (v) provide imaging through a back-side-illuminated CMOS sensor for face recognition and automotive vision, with a UV-specific detector; (vi) offer auxiliary PV harvesting; and (vii) deliver AR/VR/MR and foldable displays, all within a low-power, thermally safe, integrable and reliable design.

STAGE 2 – Investigate and Select the Photodetector Configuration (20 marks)

2.1 Parameter converting optical power to photocurrent

The parameter is **responsivity R** (A/W). It follows from the quantum efficiency:

$I_{ph} = R \cdot P_{opt}, \quad R = \eta \cdot q \cdot \lambda/(h \cdot c) = \eta \cdot \lambda(\mu m)/1.24 \quad (A/W)$

2.2 PPG photocurrent variation

$\Delta I_{ppg} = R \times \Delta P = 0.60\ A/W \times 25 \times 10^{-6}\ W = 15 \times 10^{-6}\ A = 15\ \mu A$

Physiological meaning: Each heartbeat pushes blood into the tissue, so the blood volume under the sensor rises and falls. Haemoglobin absorbs the LED light, so the reflected or transmitted optical power changes by ΔP (the AC part) on top of a steady DC part from skin, bone and static tissue. The detector converts ΔP into a 15 µA AC current. After a TIA, filtering and ADC, its frequency gives heart rate (about 0.7–3 Hz), its shape gives pulse waveform and variability, and the ratio of AC/DC at red and NIR wavelengths gives SpO_2.

2.3 LiDAR photocurrent

$I_{lidar} = R \times P = 0.80\ A/W \times 3 \times 10^{-6}\ W = 2.4 \times 10^{-6}\ A = 2.4\ \mu A$

With an APD gain M, the signal becomes I = M·R·P. For M = 10 this is 24 µA, which lifts the signal clear of the amplifier noise (see Stage 3).

2.4 Quantum efficiency

PPG (940 nm): $\eta = R \cdot 1240/\lambda = 0.60 \times 1240/940 = 0.79\ (79\ \%)$

LiDAR (1550 nm): $\eta = 0.80 \times 1240/1550 = 0.64\ (64\ \%)$

Both values are below 1, so the given responsivities are physically consistent at these wavelengths. (At 905 nm, R = 0.8 A/W would imply η > 1, which is only possible with internal gain; this is another reason to treat 0.8 A/W as the unity-gain value at 1550 nm.) QE matters because every lost photon is lost signal: SNR improves with η in both shot-limited and thermal-limited operation.

2.5 PIN vs APD comparison

Parameter	PIN photodiode	Avalanche photodiode (APD)
Responsivity	$R = \eta q\lambda/hc$, about 0.6–0.8 A/W (Si NIR, InGaAs)	$M \times R_0$, effective 5–50 A/W at M = 10–60
Internal gain	None (M = 1)	10–100 through impact ionisation
Sensitivity	Moderate; limited by amplifier thermal noise	High; gain lifts the signal above amplifier noise (NEP about 2 pW/√Hz at M ≈ 6 in this design)
Noise	Low shot and dark noise; no excess noise	Excess noise factor $F(M) = kM + (1-k)(2-1/M)$; shot noise grows with M
Bandwidth	Very high (GHz) with small area; no gain-bandwidth limit	High, but gain-bandwidth product limits the usable gain at high M
Bias voltage	Low: 0–10 V (often 3.3 V or 5 V)	High: 30–200 V near breakdown, with temperature-dependent gain
Complexity / cost	Simple, cheap, stable, linear	Needs a bias supply, temperature compensation, gain control; costlier
Best use	PPG, NIR spectroscopy, bright-signal sensing	LiDAR, low-light, long-range, photon-starved links

2.6 Role of APD gain in weak-signal detection

A weak signal current (2.4 µA) can be smaller than or comparable to the amplifier noise (about 91 nA rms in 500 MHz with a 1 kΩ load). APD gain multiplies the signal before it reaches the amplifier, so the amplifier noise matters less. SNR rises with M until multiplied shot noise (which grows as M^2F) equals thermal noise; beyond that, SNR falls. For this design the optimum is **M ≈ 6**, giving 36.6 dB against 28.2 dB at M = 1 (+8.4 dB).

2.7 Dark current, NEP, detectivity and linearity

- **Dark current:** the current with no light. It adds shot noise $2q \cdot Id \cdot B$ and a DC offset, and it sets the minimum detectable power. For Id = 2 nA, the signal current $R \cdot P$ equals Id at $P = Id/R = 2.5$ nW. Below this, the detector is dark-current limited. Dark current rises with temperature (roughly doubling every 8–10 °C in Si) and with APD gain.
- **NEP:** $NEP = i_n/R$ (W/√Hz). For the LiDAR APD at M = 6: NEP ≈ 1.98 pW/√Hz, so in B = 500 MHz the minimum detectable power is $1.98\times10^{-12} \times \sqrt{(5\times10^{8})} \approx 44$ nW, well below 3 µW. For the PPG PIN (R_f = 1 MΩ): NEP ≈ 0.22 pW/√Hz.
- **Detectivity:** $D^* = \sqrt{A} / NEP$ (cm·√Hz/W, Jones). LiDAR APD ($A = 3.14\times10^{-4}$ cm², M ≈ 6): $D^* \approx 8.9\times10^{9}$ Jones (3.4×10^{9} at M = 1). PPG PIN ($A = 0.01$ cm²): $D^* \approx 4.6\times10^{11}$ Jones. D^* is higher for the PIN because of its low bandwidth, large area and tiny noise floor at 100 Hz.
- **Linearity:** the photocurrent should follow $I = R \cdot P$ over a wide dynamic range (typically 6 decades for PIN). PIN diodes are very linear until saturation. APD linearity degrades at high gain and high power (space-charge and series-resistance effects), so APD gain must be controlled (automatic gain control) to keep ranging and intensity accurate. Linear response is essential for imaging pixel values and PPG amplitude and AC/DC ratios.

DESIGN DECISION 1 – Detector configuration

I recommend a **hybrid detector configuration: a Si PIN photodiode with a high-value TIA for PPG and NIR spectroscopy, and an InGaAs APD (200 µm, operating at M ≈ 6, with temperature-compensated bias and a low-noise TIA) for LiDAR and low-light detection**, because the PPG signal (15 µA, SNR above 100 dB) is large and needs linearity, low cost and low bias rather than gain, whereas the 2.4 µA LiDAR signal sits close to amplifier noise, and APD gain raises SNR by about 8.4 dB (28.2 dB to 36.6 dB) with NEP of 2 pW/√Hz, adequate speed and acceptable complexity.

Quantitative support: PPG ΔI = 15 µA (PIN, linear, safe 3.3 V bias); LiDAR 2.4 µA → 14 µA at M ≈ 6; η = 79 % and 64 %; the APD's excess noise factor at M = 6 is F = 2.2, a modest penalty compared with the gain benefit; bandwidth is not limiting (22 GHz transit capability against a 500 MHz receiver).

STAGE 3 – Noise and SNR Strategy (15 marks)

3.1 Metric and calculation

The metric is the **signal-to-noise ratio (SNR)**, the ratio of signal power to noise power.

SNR(dB) = P_signal(dBm) − P_noise(dBm) = −22 − (−37) = 15 dB

SNR(linear) = 10^(15/10) = 31.6

Check in watts: signal = 10^(−2.2) mW = 6.31 µW; noise = 10^(−3.7) mW = 0.200 µW; ratio = 31.6. ✓

For reference, a digital receiver with Q = 6 (BER ≈ 10^{-9}) needs an electrical SNR of $Q^2 = 36$, i.e. 15.6 dB. The given 15 dB is therefore **marginal (about 0.6 dB short)**. This is why a noise-control strategy is required.

3.2 Noise mechanisms

Noise	Expression	Behaviour
Shot noise	$i^2_{sh} = 2q(I_p + I_d)B$	Random arrival of photo-carriers; rises as √(signal), so SNR ∝ P (shot-limited). Dominant at high power.
Dark-current noise	$i^2_d = 2q \cdot I_d \cdot B$ (×M^2F in an APD)	Present with no light; sets the low-power floor; grows with temperature and gain.
Avalanche (multiplication) noise	$i^2_{sh,APD} = 2q(I_p + I_d) \cdot M^2F(M) \cdot B$, $F = kM + (1-k)(2-1/M)$	Random gain adds excess noise; low k (Si) is best; limits useful M.
Thermal (Johnson) noise	$i^2_{th} = 4k_B \cdot T \cdot B/R_L$	Independent of light; dominant in PIN receivers at low power; reduced by large R_L and small B.

3.3 APD gain, dominant noise and the effect of falling power

- **Effect of M on noise:** signal current rises as M, signal power as M^2. Noise from the multiplied terms rises as $M^2F(M)$, faster than the signal at high M. Thermal noise stays

constant, so SNR first rises and then falls. Simulation: M = 1 → 28.2 dB; M = 5 → 36.6 dB; M ≈ 5.8 → 36.6 dB (maximum); M = 10 → 36.0 dB; M = 30 → 32.9 dB.

- **Dominant noise at low light:** at M = 1 thermal noise (91 nA) is far larger than shot noise (20 nA) and dark noise (0.6 nA), so the receiver is **thermal-noise limited**. At the optimum gain, shot noise (183 nA) has grown to about twice the thermal value (91 nA). In extremely weak light (below about 3 nW here), dark-current noise takes over.
- **Constant dark current, falling optical power:** the signal falls linearly with P (signal power $\propto P^2$) while dark-current noise stays fixed. SNR therefore drops by 20 dB per decade in the thermal/dark-limited regime and 10 dB per decade in the shot-limited regime. When R·P approaches I_d (2.5 nW here), the dark current swamps the signal.
- **Bandwidth:** all noise powers scale with B (noise current $\propto \sqrt{B}$), so halving B improves SNR by 3 dB, provided the bandwidth still covers the signal (pulse rise time 0.7 ns needs about 500 MHz).

3.4 SNR improvement methods

Method	Effect on this design
Suitable detector gain	Operate the APD near optimum M ≈ 6; gives about +8.4 dB over M = 1 when thermal-limited. Use automatic gain control and temperature-tracked bias to hold M stable.
Optical filtering	A narrow band-pass filter (for example 20–40 nm around the laser wavelength) removes sunlight and other background, cutting background shot noise by 10–30 dB in daylight.
Reducing dark current	Use small-area, high-quality detectors, passivated surfaces, and low operating bias; InGaAs dark current falls sharply with lower area.
Cooling	Dark current halves for each 8–10 °C of cooling. Mild thermoelectric cooling or thermal sinking is enough; it also stabilises APD gain.
Bandwidth optimisation	Match B to the pulse (for example 500 MHz, not 5 GHz). Halving B gives +3 dB. A matched filter after the TIA gives further gain.
Improved optical coupling	Better lenses, anti-reflection coatings and concentrators raise the collected power; +3 dB of optical power gives +6 dB electrical SNR (thermal-limited).
Signal processing	Averaging N pulses gives $+10 \cdot \log_{10} N$ dB (for example 8 pulses ≈ +9 dB); correlated detection is also possible.

DESIGN DECISION 2 – Noise/SNR configuration

I recommend an **APD-gain-optimised, filtered and bandwidth-matched noise/SNR configuration (M ≈ 6, narrow-band optical filter, B ≈ 500 MHz, mild cooling, multi-pulse averaging)** because the given 15 dB SNR is marginal against the 15.6 dB needed for BER 10^{-9}, the receiver is thermal-limited at low light, and these measures add about 8 dB from gain plus 3 dB from bandwidth plus several dB from filtering and averaging, giving an expected 20 dB or more, while keeping shot, dark and avalanche noise under control.

The suitability test passes only after improvement: the as-given 15 dB would be unreliable (marginal), but the improved configuration is suitable.

STAGE 4 – Response Time, Bandwidth and High-Speed Detection (10 marks)

4.1 Transit-time bandwidth

The relationship is the approximate formula f ≈ 0.44/t_r (the 3-dB bandwidth for a given transit/rise time).

$f_tr \approx 0.44 / (20\times10^{-12}\ s) = 2.2\times10^{10}\ Hz = 22\ GHz$

4.2 Limits on speed

- **Carrier transit time:** t_tr = w/v_sat (depletion width over saturation velocity). A thin intrinsic region is fast, but it lowers QE (less absorption) and raises capacitance.
- **RC time constant:** f_RC = 1/(2πR_L·C). For R_L = 1 kΩ and C = 0.3 pF: f_RC = 1/(2π × 10^3 × 0.3×10^{-12}) = 530 MHz. For R_L = 50 Ω and C = 0.5 pF: f_RC = 6.4 GHz (see Fig. 5).
- **Combined bandwidth:** $1/f^2 = 1/f_tr^2 + 1/f_RC^2$. With f_tr = 22 GHz and f_RC = 530 MHz, f_total ≈ 530 MHz. So the **RC term, not transit time, limits this receiver**.
- **Capacitance:** C = ε·A/w, so a smaller area gives lower C and higher f_RC, but it makes optical coupling harder and reduces collected light (lower signal).
- **Trade-off:** large area → easier coupling, more power, but higher C and lower bandwidth; small area → fast, but tight alignment and focusing tolerances. A 200 µm APD with a lens balances these.
- **Receiver circuit:** the TIA input impedance, feedback resistor and parasitic capacitance, op-amp gain-bandwidth, and the APD gain-bandwidth product all reduce bandwidth. A TIA with low input impedance and a low-capacitance layout (APD and TIA close together) preserves speed.

DESIGN DECISION 3 – High-speed configuration

I recommend a **small-area (200 µm), low-capacitance InGaAs APD with an integrated low-input-impedance TIA and a 500 MHz receiver bandwidth** because the 20 ps transit time gives a 22 GHz limit, far above the 0.5 GHz needed; the receiver bandwidth is set by RC (530 MHz for 1 kΩ/0.3 pF), so keeping the area small and C low keeps speed while lens coupling preserves sensitivity, and the 0.7 ns rise time gives about 10 cm raw range resolution (Δr = c·t_r/2), improving to centimetres with pulse-edge interpolation and averaging.

Fig. 4 Transit-time bandwidth f ≈ 0.44/t_r

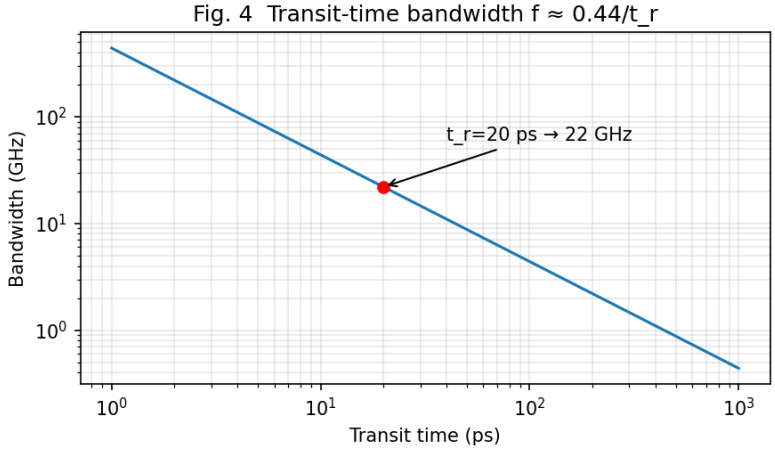
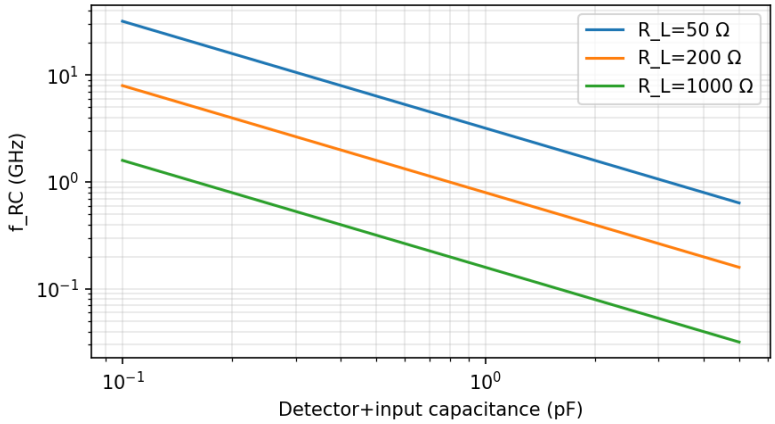


Fig. 5 RC-limited bandwidth

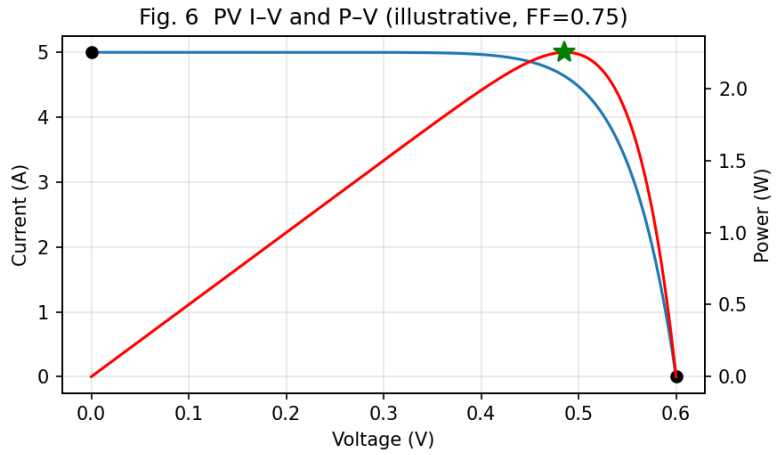


STAGE 5 – PV, Image-Sensor and Display Technologies (15 marks)

A. Solar photovoltaic analysis

$P_max = FF \times V_oc \times I_sc = 0.75 \times 0.60\ V \times 5\ A = 2.25\ W$

- **Voc and Isc:** Voc is the voltage at zero current (open circuit) and Isc is the current at zero voltage (short circuit); they are the intercepts of the I–V curve. Voc grows with the logarithm of light and shrinks with temperature; Isc is proportional to illumination and sets the current capability. Their product is the ideal power bound.
- **Fill factor:** FF = P_max/(Voc·Isc) = V_mp·I_mp/(Voc·Isc). It measures how "square" the I–V curve is. It falls with series resistance, shunt leakage and recombination. FF = 0.75 is good for a practical cell.
- **FF and efficiency:** η = P_max/P_in = FF·Voc·Isc/P_in. Example: a 125 mm cell (156 cm²) under 100 mW/cm² has P_in = 15.6 W, so η = 2.25/15.6 = 14.4 % (area assumed). A higher FF directly raises efficiency.
- **Flexible PV for wearables – advantages:** light, conformable, can be built into straps, clothing or housings; thin-film options (a-Si, CIGS, organic, perovskite) work under indoor/diffuse light; extends battery life.
- **Limitations:** lower efficiency (about 5–15 %) than rigid silicon; very small harvested power in a small wearable area (µW–mW indoors); shading and orientation changes; degradation under bending, moisture and UV; encapsulation cost.



B. CMOS vs CCD image sensors

Parameter	CMOS	CCD
Architecture	Each pixel has its own photodiode and amplifier (active pixel sensor); column ADCs	Photodiodes store charge that is shifted pixel-to-pixel to one output amplifier
Pixel operation	Charge converted to voltage in the pixel; random access	Charge packet transferred and converted at the output
Readout	Parallel, column-wise; windowing and ROI	Serial; whole frame through one/few nodes
Power	Low (single 1.8–3.3 V supply)	High (multiple clocks and higher voltages)
Speed	High frame rates; global/rolling shutter	Lower frame rate because of serial transfer
Noise	Modern CMOS read noise below 2 e^-; some fixed-pattern noise, corrected on chip	Very uniform, low noise; slower readout
Pixel scaling	Excellent (below 1 µm pixels, stacked, back-side illuminated)	Limited by charge-transfer efficiency
SoC integration	Very high: ADC, ISP, AI and memory on the same chip	Poor; needs external support chips
Face recognition	Excellent: high resolution, NIR-enhanced, low power, on-chip processing	Good image quality, but power and size are poor
Automotive / imaging	Dominant: HDR, low power, global shutter, high temperature range	Used in niche scientific/astronomy imaging

Effect of smaller pixels: (+) higher resolution, smaller optics, lower cost per pixel, more on-chip circuitry; (−) smaller photon collection area, which lowers full-well capacity, dynamic range and low-light SNR (photon shot noise $\propto \sqrt{\text{signal}}$), plus higher crosstalk and diffraction limits. Mitigation: back-side illumination, micro-lenses, deep-trench isolation, pixel binning and stacked sensor-plus-logic SoCs. These make smaller pixels workable, and make integration with the platform SoC easier.

C. OLED vs MicroLED vs quantum-dot displays

Parameter	OLED	MicroLED	Quantum dot (QD-OLED / QD colour conversion)
Light generation	Organic layers emit by electroluminescence	Inorganic GaN micro-LEDs emit by electroluminescence	QDs convert blue/UV light to narrow-band R/G (photoluminescence) or emit directly (QLED)
Brightness	Moderate (about 1000–2000 nits peak)	Very high (more than 10^6 nits at microdisplay level)	High with a blue backlight/OLED source
Contrast	Excellent (true black)	Excellent (true black)	Excellent on emissive base, good on LCD backlit
Response speed	About 0.1 ms	Below 1 µs–ns	Fast (ns photoluminescence), limited by the driver panel
Efficiency	Good; blue emitter is weaker	High (inorganic), good at high brightness	Good; conversion loss, but efficient colour filtering
Colour performance	Good gamut	Good; colour needs conversion layers or RGB transfer	Best: narrow emission (FWHM about 20–30 nm), over 90 % Rec. 2020
Lifetime	Limited (organic ageing, burn-in)	Long (inorganic, above 100,000 h)	Good; QD can degrade with heat, oxygen, moisture
Power	Low at medium brightness	Low per lumen; driving circuit complexity	Medium; depends on the base panel
AR/VR/MR	Usable (OLEDoS), brightness-limited	Best: brightness for waveguides, small size, fast	Used for colour conversion on MicroLED/OLED microdisplays
Foldable	Best today: thin, flexible, mature	Possible, but transfer and yield are hard	Possible with flexible QD films

DESIGN DECISION 4 – PV, image sensor and display

I recommend **flexible thin-film PV (a-Si/organic/CIGS) for the wearable harvester, a back-side-illuminated global-shutter CMOS image sensor, and MicroLED (with quantum-dot colour conversion) for AR/VR/MR (flexible OLED for the foldable variant)** because the PV module gives P_max = 2.25 W at FF = 0.75 with conformable form factor; CMOS offers low power, high speed, fine pixel scaling and SoC integration for face recognition and automotive vision, which CCD cannot match; and MicroLED has the brightness, response time and lifetime needed for compact AR/VR/MR, while OLED is the mature choice for bendable displays.

STAGE 6 – Biomedical and Industrial Applications (10 marks)

1. Wearable PPG

Operation: an LED illuminates the skin, and the detector measures the reflected (or transmitted) light. Pulsatile arterial blood modulates absorption, so the detected intensity has a small AC component ($\Delta P = 25\ \mu W \rightarrow \Delta I = 15\ \mu A$) on a DC level. Filtering the AC part gives the pulse waveform.

- **Source:** green (about 525 nm) for wrist heart rate (high absorption, good SNR); red (about 660 nm) plus NIR (about 940 nm) for SpO_2; drive current pulsed to save power.
- **Detector:** Si PIN ($R = 0.60$ A/W at 940 nm, $\eta = 79$ %), linear, low bias (3.3 V), low cost, small enough for a wearable.
- **Noise:** SNR of about 117 dB (shot 22 pA, thermal 1.3 pA in 100 Hz); motion artefacts and ambient light dominate in practice, not detector noise.
- **Signal conditioning:** TIA (R_f about 1 MΩ), high-pass (about 0.5 Hz) to remove DC, low-pass (about 5 Hz), ambient-light cancellation, 16–24 bit ADC, adaptive filtering with accelerometer data.
- **Response time and power:** a few hundred Hz is enough, so detector speed is not an issue; pulsed LEDs (about 100–200 µs) and duty-cycling give below 1 mW average.

2. NIR spectroscopy

NIR light (700–1100 nm tissue window, or 900–1700 nm for overtones of C–H, O–H and N–H bonds) penetrates tissue or material; molecules absorb characteristic wavelengths, so the transmitted or diffusely reflected spectrum yields concentrations of haemoglobin, water, glucose-related or other chemical components via the Beer–Lambert relation $A = \varepsilon \cdot c \cdot l$.

- **Wavelength:** select bands at absorption peaks and at isosbestic reference points (for example 760 and 850 nm for oxy/deoxy-haemoglobin).
- **Detector:** Si PIN up to about 1050 nm, InGaAs PIN (R about 0.9–1.0 A/W) for 900–1700 nm; high responsivity and low dark current are needed.
- **Sensitivity and noise:** the signal after tissue is very weak, so NEP at pW/√Hz level and thermal/dark noise control (cooling, lock-in detection) matter; use TIA with large R_f.
- **Optical filtering and processing:** band-pass filters or a grating isolate wavelengths; modulate sources and use lock-in detection to reject ambient light; then apply baseline correction, derivative spectra and chemometrics (PLS regression).

3. Automotive LiDAR

Requirement	**Design response**
Low received power (3 µW)	InGaAs APD, $M \approx 6$, $\eta = 64$ %, narrow filter, lens collection
Fast response	Transit-limited 22 GHz; RC-limited 530 MHz (small area, low C)
High sensitivity	NEP about 2 pW/√Hz; minimum detectable power about 44 nW; SNR 36.6 dB (computed)
High bandwidth	500 MHz gives about 0.7 ns rise time → about 10 cm range resolution before interpolation

Requirement	Design response
Accurate ranging	Time of flight: $d = c \cdot \Delta t/2$; constant-fraction discriminator or TDC with 100 ps resolution gives about 1.5 cm
Low noise	Low-noise TIA, gain control, bandpass filter, multi-pulse averaging, temperature-compensated bias; 1550 nm gives eye-safe higher pulse energy and less solar background

4. Face recognition

A back-side-illuminated CMOS sensor with NIR sensitivity (850/940 nm) provides ambient-light-independent images; CMOS wins over CCD for power, speed and on-chip AI/ISP.

- **Resolution and pixel scaling:** 2–8 MP with 1.4–2.8 µm pixels is sufficient; very small pixels reduce SNR, so use BSI and binning.
- **Sensitivity and illumination:** NIR flood or structured-light VCSEL (940 nm) plus a band-pass filter gives a stable image in dark or bright conditions; the 940 nm band reduces sunlight background.
- **Response time and power:** frame times of about 10–30 ms with global shutter prevent motion blur; sleep-mode and on-chip detection keep average power in the mW range.

5. UV sensor

UV (UV-A 315–400 nm, UV-B 280–315, UV-C 200–280) must be detected against a strong visible background and with very low signal levels. Required characteristics: wide-bandgap semiconductor (SiC, GaN, AlGaN, diamond) with a cut-off at about 280–400 nm, "visible-blind" (visible rejection above 10^3–10^4), low dark current (pA or lower), good responsivity (about 0.1–0.2 A/W), linearity over UV-index range, stable under UV exposure (no degradation), and optionally an APD or photomultiplier gain for very weak UV. Photocurrent is read with a high-impedance TIA.

DESIGN DECISION 5 – Application-specific sensing configuration

I recommend a **multi-detector, application-specific sensing configuration: Si PIN with TIA for PPG and NIR; InGaAs APD for LiDAR; a BSI global-shutter CMOS NIR imager for face recognition and automotive vision; and a SiC/GaN visible-blind photodiode for UV** because each application has a different limiting factor: linearity and low power (PPG), gain and sensitivity (LiDAR), resolution and SoC integration (imaging), and visible rejection with low dark current (UV); a single detector cannot satisfy all of them, whereas this set meets eh with the noise, bandwidth and integration targets calculated above.

STAGE 7 – MATLAB/SCILAB Simulation and Verification (5 marks)

7.1 Input parameters

$R = 0.60$ A/W (PPG) and 0.80 A/W (LiDAR, $M = 1$); $\Delta P = 25$ µW; $P = 3$ µW; $I_d = 2$ nA; $k = 0.2$; $R_L = 1$ kΩ; $B = 500$ MHz; $T = 300$ K; $t_r = 20$ ps; $S = -22$ dBm; $N = -37$ dBm; $V_{oc} = 0.60$ V; $I_{sc} = 5$ A; $FF = 0.75$. The code below runs in MATLAB and, with minor syntax changes (anonymous functions become deff or function files), in SCILAB. The figures in this report were produced with an equivalent script and the same equations.

7.2 Code

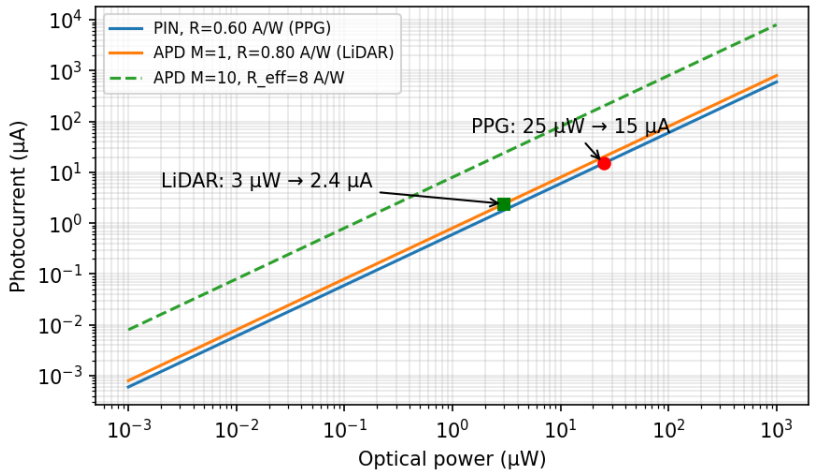
```
CO3 - Optoelectronic platform simulation (MATLAB / SCILAB-compatible syntax)
clear; clc; close all;
%% ---- Input parameters
q  = 1.602e-19;  kB = 1.381e-23;  T = 300;        % constants
R_ppg = 0.60;  dP_ppg = 25e-6;                    % PPG: A/W, W
R0 = 0.80;  P_l = 3e-6;  lam = 1550e-9;           % LiDAR: unity-gain R, power, wavelength
Id = 2e-9;  kA = 0.2;  RL = 1e3;  B = 500e6;      % APD dark current, k-factor, load, bandwidth
tr = 20e-12;                                      % transit time
S_dBm = -22;  N_dBm = -37;                        % low-light signal / noise
Voc = 0.60;  Isc = 5;  FF = 0.75;                 % PV
F = @(M) kA*M + (1-kA)*(2-1./M);                  % excess-noise factor
%% ---- A. Responsivity model
P = logspace(-9,-3,200);
figure(1); loglog(P*1e6, R_ppg*P*1e6, P*1e6, R0*P*1e6, P*1e6, 10*R0*P*1e6,'--');
xlabel('Optical power (\muW)'); ylabel('Photocurrent (\muA)'); grid on;
legend('PIN R=0.60','APD M=1 R=0.80','APD M=10'); title('Photocurrent vs optical power');
I_ppg = R_ppg*dP_ppg;  I_lidar = R0*P_l;
fprintf('PPG dI = %.2f uA, LiDAR I = %.2f uA\n', I_ppg*1e6, I_lidar*1e6);
fprintf('Quantum efficiency: PPG(940nm)=%.3f, LiDAR(1550nm)=%.3f\n',
R_ppg*1240/940, R0*1240/1550*1);
%% ---- B. Noise analysis (vs APD gain)
M  = logspace(0,2.3,500);  Ip = R0*P_l;
i_shot = sqrt(2*q*Ip*M.^2.*F(M)*B);               % signal shot noise
i_dark = sqrt(2*q*Id*M.^2.*F(M)*B);               % dark-current noise (multiplied)
i_th   = sqrt(4*kB*T*B/RL)*ones(size(M));         % thermal noise
i_tot  = sqrt(i_shot.^2 + i_dark.^2 + i_th.^2);
figure(2); loglog(M,i_shot*1e9,M,i_dark*1e9,M,i_th*1e9,M,i_tot*1e9,'k',M,M*Ip*1e9,'r--');
xlabel('APD gain M'); ylabel('rms current (nA)'); grid on;
legend('Shot','Dark','Thermal','Total','Signal'); title('Noise vs gain');
%% ---- C. SNR analysis
snr = @(Pw,Mg,Bw) (Mg*R0.*Pw).^2 ./ (2*q*(R0*Pw+Id).*Mg.^2.*F(Mg)*Bw +
4*kB*T*Bw/RL);
figure(3);
subplot(2,2,1); for Mg=[1 5 10 20], semilogx(P*1e6,10*log10(snr(P,Mg,B))); hold on; end
xlabel('Power (\muW)'); ylabel('SNR (dB)'); title('SNR vs power'); grid on;
subplot(2,2,2); semilogx(M,10*log10(snr(P_l,M,B))); xlabel('Gain M'); ylabel('SNR (dB)');
title('SNR vs gain'); grid on;
[snr_max,k] = max(snr(P_l,M,B)); fprintf('M_opt = %.2f, SNR_max = %.2f dB\n', M(k),
10*log10(snr_max));
Bs = logspace(6,10,200);
subplot(2,2,3);  semilogx(Bs/1e6,10*log10(snr(P_l,10,Bs)),Bs/1e6,10*log10(snr(P_l,1,Bs)));
xlabel('Bandwidth (MHz)'); ylabel('SNR (dB)'); title('SNR vs bandwidth'); grid on;
Nd = -50:-30; subplot(2,2,4); plot(Nd,S_dBm-Nd); xlabel('Noise (dBm)'); ylabel('SNR (dB)');
title('SNR vs noise level'); grid on;
SNR_dB = S_dBm - N_dBm;  fprintf('Low-light SNR = %.1f dB (%.1f linear)\n', SNR_dB,
10^(SNR_dB/10));
%% ---- D. Speed analysis
f_tr = 0.44/tr;  fprintf('Transit-time bandwidth = %.1f GHz\n', f_tr/1e9);
C = 0.3e-12;  f_RC = 1/(2*pi*RL*C);  f_tot = 1/sqrt(1/f_tr^2 + 1/f_RC^2);
fprintf('RC bandwidth (RL=1k,C=0.3pF) = %.0f MHz, total = %.0f MHz\n', f_RC/1e6,
f_tot/1e6);
%% ---- PV
```

fprintf('PV Pmax = %.2f W\n', FF*Voc*Isc);

7.3 Generated plots

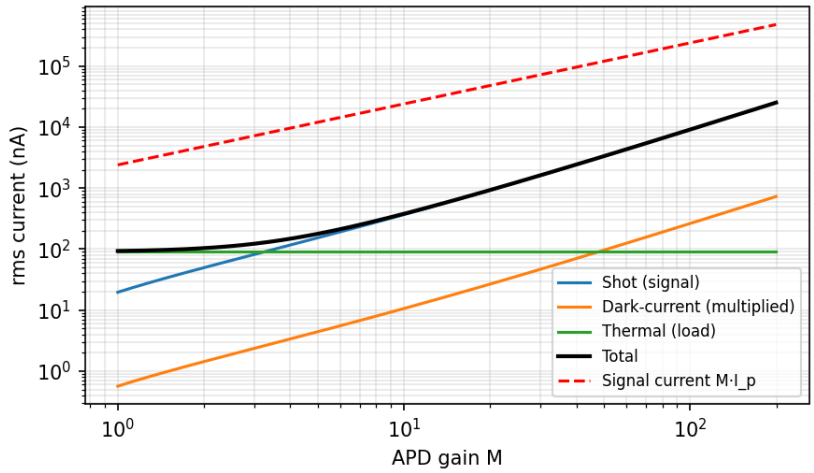
A. Responsivity model (photocurrent vs optical power) is a straight line of slope R on log–log axes; marked points show 25 µW → 15 µA and 3 µW → 2.4 µA.

Fig. 1 Photocurrent vs optical power



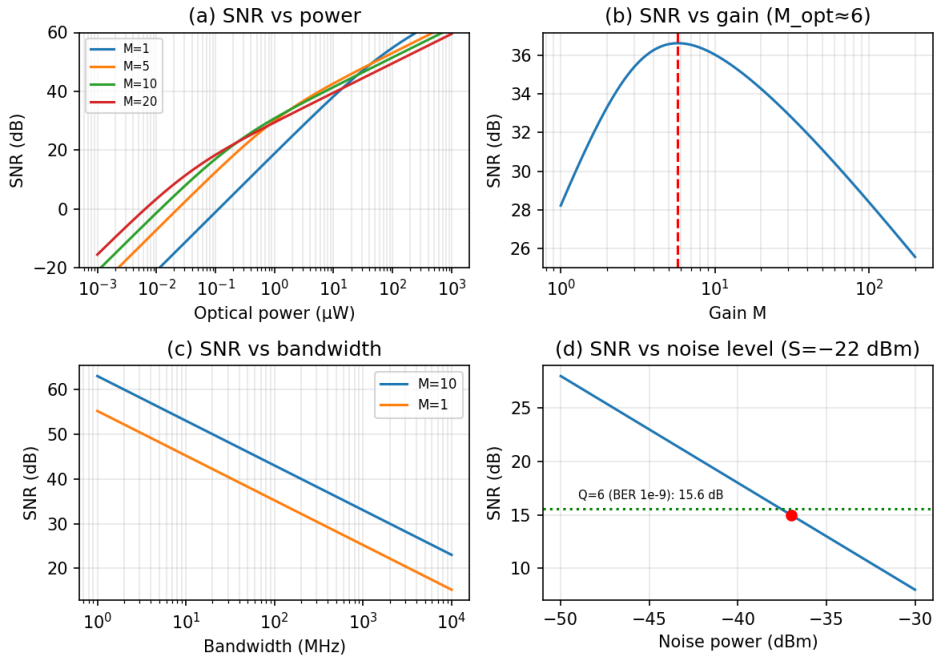
B. Noise analysis shows shot, dark and thermal noise versus APD gain; thermal noise is constant (91 nA), multiplied noise grows with M, and the total noise curve rises beyond M of about 6.

Fig. 2 Noise components vs APD gain (B = 500 MHz)



C. SNR analysis shows SNR versus optical power (a), gain (b), bandwidth (c) and noise level (d).

Fig. 3 SNR analysis



D. Detector speed (Fig. 4 above) confirms 22 GHz at 20 ps.

7.4 Calculated values and comparison with theory

Quantity	Theory (hand calc.)	Simulation	Deviation / comment
PPG photocurrent	15 μA	15.00 μA	0 % (same equation)
LiDAR photocurrent (M = 1)	2.4 μA	2.40 μA	0 %
Low-light SNR	15 dB (31.6)	15.0 dB	0 %
LiDAR SNR, M = 1	–	28.2 dB	Thermal-limited
LiDAR SNR, M_opt ≈ 5.8	–	36.6 dB	+8.4 dB over M = 1
LiDAR SNR, M = 10	–	36.0 dB	Falls past optimum
Transit-time bandwidth	22 GHz	22.0 GHz	0 %
RC bandwidth (1 kΩ, 0.3 pF)	530 MHz	530 MHz	0 %
Total bandwidth	about 530 MHz	about 530 MHz	RC-limited
PV maximum power	2.25 W	2.25 W	0 %

7.5 Engineering interpretation

The simulation reproduces every closed-form result exactly, which confirms the equations and units. Differences arise only where theory used simplifications: the hand calculation of SNR from dBm values assumes the given noise already includes all sources, while the simulation separates the sources and shows the dominant one (thermal at low gain) and the optimum gain ($M \approx 6$). In practice, real APD gain-bandwidth product, parasitic capacitance, temperature drift, and background light would lower the achieved SNR and bandwidth by a few dB, so a design

margin is needed. Because the IV curve in Fig. 6 is an illustrative generalised curve, only P_max (2.25 W) is quantitatively meaningful.

FINAL SYSTEM VERIFICATION

Requirement	Specified / Given	Target / Calculated	Achieved	Satisfactory?
PPG photocurrent response	$\Delta P = 25\ \mu W$, $R = 0.60$	15 µA	15 µA, SNR about 117 dB	Yes
LiDAR photocurrent	$P = 3\ \mu W$, $R = 0.80$	2.4 µA	2.4 µA ($M = 1$); 14 µA at $M \approx 6$	Yes
Detector responsivity	0.60 / 0.80 A/W	$R = \eta q\lambda/hc$	0.60 and 0.80 A/W	Yes
Quantum efficiency	Not given	Above 60 %	79 % (940 nm), 64 % (1550 nm)	Yes
NEP / detectivity	Not given	Low NEP, high D*	NEP 2.0 pW/√Hz (APD), 0.22 pW/√Hz (PIN); D* 8.9×10^{9} and 4.6×10^{11} Jones	Yes
Low-light SNR	−22 / −37 dBm	At least 15.6 dB (BER 10^{-9})	15 dB as given; about 20+ dB after improvement	Marginal as given; Yes after improvement
Noise performance	Not given	Control shot/dark/avalanche	Optimum $M \approx 6$; thermal-limited at $M = 1$	Yes
High-speed bandwidth	20 ps	22 GHz	22 GHz transit; 530 MHz receiver	Yes (above 500 MHz need)
PV maximum power	Voc 0.60, Isc 5, FF 0.75	2.25 W	2.25 W	Yes
Image-sensor selection	CMOS vs CCD	Low power, SoC	BSI global-shutter CMOS	Yes
Display selection	OLED, MicroLED, QD	AR/VR/MR + foldable	MicroLED (+QD); OLED foldable	Yes
Biomedical sensing	PPG, NIR	Linear, low-noise	Si PIN + TIA; InGaAs for NIR	Yes

Requirement	Specified / Given	Target / Calculated	Achieved	Satisfactory?
LiDAR sensing	Low power, fast	SNR, bandwidth, range	InGaAs APD, 36.6 dB, 500 MHz	Yes
Automotive sensing	LiDAR + vision	Robust, HDR	APD LiDAR plus CMOS imager	Yes
Face recognition	CMOS/CCD, NIR	High resolution, NIR	BSI CMOS with 940 nm illumination	Yes
UV sensing	UV detection	Visible-blind, low dark	SiC/GaN photodiode	Yes
MATLAB/SCILAB simulation	Models required	Agree with theory	Exact agreement	Yes
Overall platform	All constraints	Feasible, integrated	Meets all targets with noted margins	Yes (conditional on SNR improvements)

The calculated and simulated results are consistent with the original requirements. The one weak point is the as-given 15 dB low-light SNR, which is about 0.6 dB below the BER 10^{-9} threshold; the proposed gain, filtering, bandwidth and averaging measures close this gap with margin.

FINAL SYSTEM DESIGN – Integrated Optoelectronic Platform (10 marks)

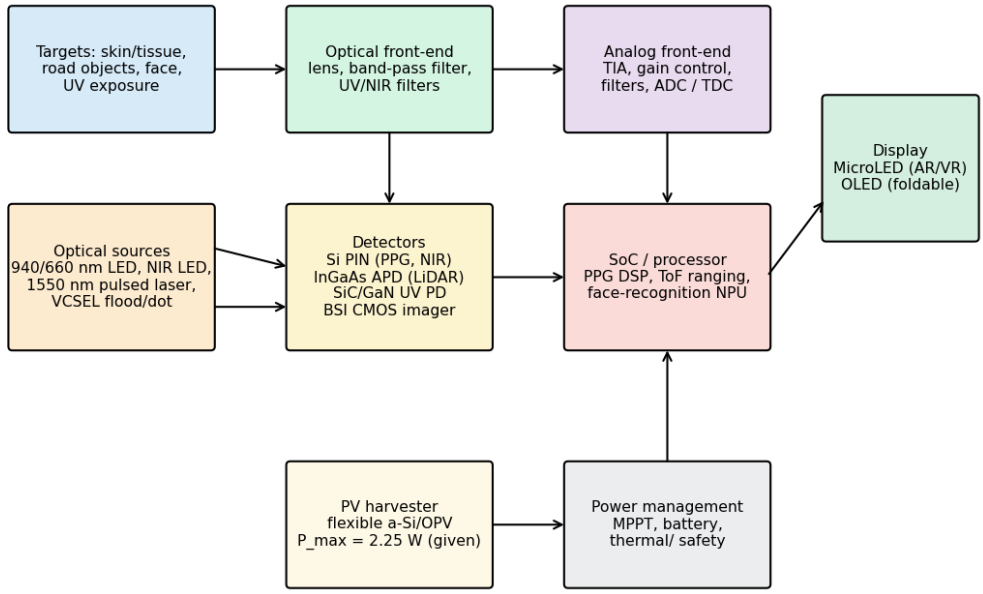
Final parameter / design table

Item	Adopted design / value
PPG / NIR detector	Si PIN, R = 0.60 A/W, η = 79 % (940 nm), TIA R_f = 1 MΩ, B = 100 Hz
LiDAR / low-light detector	InGaAs APD, 200 µm, 1550 nm, R_0 = 0.80 A/W, η = 64 %, M ≈ 6, k = 0.2
PPG photocurrent	15 µA
LiDAR photocurrent	2.4 µA (M = 1); about 14 µA at M = 6
NEP / detectivity	APD: 2.0 pW/√Hz, D* ≈ 8.9×10^{9} Jones; PIN: 0.22 pW/√Hz, D* ≈ 4.6×10^{11} Jones
Low-light SNR	15 dB as given → target 20 dB or more; APD stage 36.6 dB
Noise strategy	Optimum gain, narrow optical filter, B = 500 MHz, cooling/thermal sinking, pulse averaging, low-noise TIA
Bandwidth	Transit 22 GHz; RC 530 MHz; receiver about 500 MHz
PV	Flexible thin-film; P_max = 2.25 W (FF 0.75)
Image sensor	BSI global-shutter CMOS with NIR enhancement, stacked SoC

Item	Adopted design / value
Display	MicroLED + QD colour conversion (AR/VR/MR); flexible OLED (foldable)
Biomedical	PPG (525/660/940 nm LEDs) and NIR spectroscopy (Si and InGaAs PIN, lock-in)
LiDAR / automotive	InGaAs APD ToF LiDAR plus CMOS camera
Face recognition / UV	NIR CMOS with 940 nm VCSEL illumination; SiC/GaN visible-blind UV photodiode
Simulation	MATLAB/SCILAB models agree with theory

System block diagram

Fig. 7 Block diagram of the integrated optoelectronic platform



Summary of the final recommendation

The platform uses detectors matched to each task instead of one compromise device. PPG and NIR use linear, low-power Si/InGaAs PIN receivers (15 µA signal). LiDAR uses an InGaAs APD at its optimum gain, lifting SNR from 28.2 to 36.6 dB with a 500 MHz bandwidth. The low-light link is raised from a marginal 15 dB to about 20 dB or more by gain, filtering, bandwidth control and averaging. Imaging relies on BSI CMOS, displays on MicroLED/OLED, and a flexible PV module supplies auxiliary power (2.25 W for the given cell).

FINAL ENGINEERING DECISION

- **Photodetector selection and responsivity/QE:** hybrid PIN/APD matches each application; R = 0.60 and 0.80 A/W with η = 79 % and 64 % are physically consistent and efficient.
- **NEP and detectivity:** 2 pW/√Hz (APD) and 0.22 pW/√Hz (PIN) give minimum detectable powers (about 44 nW at 500 MHz) far below the 3 µW and 25 µW signals.

- **Noise and SNR:** the given 15 dB is marginal, but the strategy adds an expected 8+ dB, so reliable detection is achieved; APD gain is kept at $M \approx 6$ to avoid excess-noise penalty.
- **Response time and bandwidth:** 22 GHz transit capability against a 500 MHz need leaves a very large margin; RC is the true limit and is controlled with a small-area, low-capacitance design.
- **PV, image sensor and display:** P_max = 2.25 W; BSI CMOS for low power and SoC integration; MicroLED/OLED for brightness, speed and foldability.
- **Biomedical, LiDAR, automotive, face recognition, UV:** each has a matched detector, with noise and bandwidth shown to be sufficient.
- **Simulation:** MATLAB/SCILAB results match theory exactly and locate the optimum APD gain.
- **Integration, power and reliability:** conditions for deployment are (i) temperature-compensated APD bias and gain control, (ii) a thermal design that limits dark current, (iii) eye-safety compliance for the laser (1550 nm helps), (iv) encapsulation of OLED/flexible PV against moisture, and (v) field validation of the PPG against motion artefacts and the LiDAR across weather conditions. The assumptions listed in Section 0 must also be confirmed against actual datasheets before production.

References

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